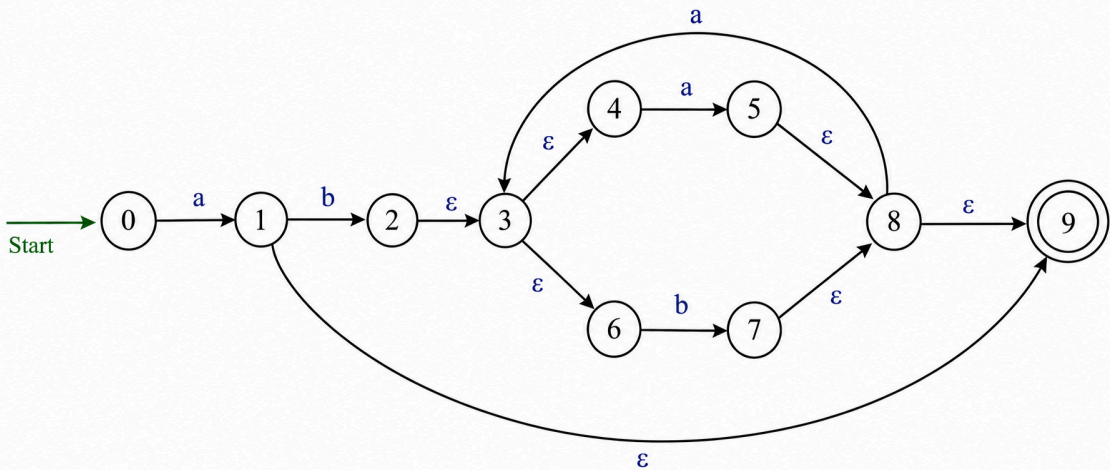


Hope Foundation's	
International Institute of Information Technology (I2IT), Pune	
DEPARTMENT OF COMPUTER ENGINEERING	
Academic Year 2026-27 Semester I	
THEORY QUESTION BANK	
Class: TE CE Div-A, B , C	
Course Name: Theory of Computations	
Que No.	Question
Unit 1 (CO303.1)	
Q1.	Define a Finite Automaton. Explain the components of a DFA with a neat diagram.
Q2.	List the differences between DFA, NFA, and ϵ -NFA. Provide examples for each.
Q3.	Convert the following Mealy machine to an equivalent Moore machine: States = {S0, S1}, Input: {0,1}, Transitions: S0 $\rightarrow$ 0/1 $\rightarrow$ S1, S1 $\rightarrow$ 1/0 $\rightarrow$ S0.
Q4.	Define Moore and Mealy machines. List key differences with example outputs.
Q5.	What is the language accepted by a Finite Automaton? Describe with an example.
Q6.	Construct a DFA that accepts the set of strings over {0,1} that contain an even number of 0s. Show transition table and final state.
Q7.	Given an NFA with transitions: $\delta(q_0, a) = \{q_0, q_1\}$, $\delta(q_1, b) = \{q_2\}$, $\delta(q_2, a) = \{q_2\}$, draw the transition diagram and find an equivalent DFA.
Q8.	Draw FA for the following language over i) Number of 1's is multiple of 3 ii) Number of 1's is not multiple of 3
Q9.	Convert the given ϵ -NFA to equivalent DFA. $\delta(q_0, \epsilon) = \{q_1\}$, $\delta(q_1, a) = \{q_2\}$, $\delta(q_2, b) = \{q_3\}$, $\delta(q_3, a) = \{q_1\}$.
Q10.	Construct a Mealy machine to output 1 when the input string ends in "ab". Output 0 otherwise. Show state transitions and output.
Q11.	Construct a DFA to accept the following i) The set of all strings over {a,b} ending with ab. ii) The set of all strings over {0,1} having an even number of 0's and odd number of 1's.

Q12.	Construct Moore machine for the following $Q = \{q_0, q_1, q_2, q_3\}$, $\Sigma = \{a, b\}$, $\Delta = \{0, 1\}$ and δ is shown as given below. <table><tr><td>δ</td><td>a</td><td>b</td><td>λ</td></tr><tr><td>q0</td><td>q3</td><td>q2</td><td>0</td></tr><tr><td>q1</td><td>q1</td><td>q0</td><td>0</td></tr><tr><td>q2</td><td>q2</td><td>q3</td><td>1</td></tr><tr><td>q3</td><td>q0</td><td>q1</td><td>0</td></tr></table>	δ	a	b	λ	q0	q3	q2	0	q1	q1	q0	0	q2	q2	q3	1	q3	q0	q1	0
δ	a	b	λ																		
q0	q3	q2	0																		
q1	q1	q0	0																		
q2	q2	q3	1																		
q3	q0	q1	0																		
Q13.	Design a DFA for the language: $L = \{w \mid w \text{ contains at most one '1'}\}$. Clearly label states and transitions.																				
Q14.	Explain the step-by-step minimization of the DFA defined by: States = $\{A, B, C, D\}$, start = A, final = C, transitions: $A \rightarrow 0 \rightarrow B$, $A \rightarrow 1 \rightarrow C$, $B \rightarrow 0 \rightarrow D$, $B \rightarrow 1 \rightarrow C$, $C \rightarrow 0 \rightarrow C$, $C \rightarrow 1 \rightarrow C$, $D \rightarrow 0 \rightarrow D$, $D \rightarrow 1 \rightarrow C$.																				
Q15.	Convert the given ϵ-NFA to DFA 																				
Unit 2 (CO303.2)																					
Q1.	Define Regular Expression. List and explain the operators used in RE with suitable examples.																				
Q2.	What is the precedence of operators in regular expressions? Write the standard order of evaluation with examples.																				
Q3.	List and explain algebraic laws of regular expressions. Illustrate any 3 laws using expressions.																				
Q4.	Define the Pumping Lemma. State and explain its use in proving languages are not regular.																				
Q5.	What are the closure properties of regular languages? Explain with examples.																				
Q6.	Construct a Regular Expression for the language: $L = \{w \mid w \text{ begins with 'a' and ends with 'b', over } \Sigma = \{a, b\}\}$.																				

Q7.	Convert the following regular expression into NFA: $R = (a + b)^*ab$. Show step-by-step transitions and draw the NFA.
Q8.	Prove that the language $L = \{a^n b^n \mid n \geq 0\}$ is not regular using the Pumping Lemma.
Q9.	Construct Regular expression for following DFA using Arden's theorem. <pre> graph LR Start((Start)) --> q1((q1)) q1 -- a --> q1 q1 -- a --> q2((q2)) q2 -- b --> q1 q2 -- b --> q2 q2 -- a --> q3(((q3))) q3 -- a --> q2 </pre>
Q10.	Write RE for following - Obtain a regular expression to accept strings of a's and b's whose second symbol from the right is 'a' - Obtain a regular expression to accept strings of a's and b's starting with 'a' and ending with 'b' - Obtain a regular expression to accept strings of a's and b's for the language $L = \{a^{2n} b^{2m} \mid n \geq 0, m \geq 0\}$ - Obtain a regular expression to accept strings of a's and b's for the language $L = \{a^n b^m \mid n+m \text{ is even}\}$
Q11.	Convert the following regular expression to DFA: $R = ab(a + b)$. Show all intermediate steps.
Q12.	Convert the following RE to ϵ -NFA and find the ϵ -closure of all the states and corresponding DFA. $(0+1)^*. 1.(0+1)$
Q13.	Consider the two RE $r = 0^* + 1^*$, $s = 01^* + 10^* + 1^*0 + (0^*1)^*$ - Find the string corresponding to r but not to s. - Find the string corresponding to s but not to r. - Find the string corresponding to both r & s. - Find the string corresponding to neither r nor s.
Q14.	Given the DFA with: States = $\{S0, S1, S2\}$, Start = S0, Final = S2, Transitions: $S0 \rightarrow a \rightarrow S1$, $S1 \rightarrow b \rightarrow S2$, $S2 \rightarrow a \rightarrow S2$, $S2 \rightarrow b \rightarrow S2$ Use the state elimination method to find an equivalent RE.
Q15.	Design a regular expression for $L = \{w \mid w \text{ has an even number of a's and any number of b's}\}$.